Where the Error of the Factorised Cumulant Chain Lives, and What It Costs

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We locate where the error of a factorised cumulant chain for deep random ReLU MLPs is made, and what removing it would cost. The closures are accurate given true inputs; the error is input error made at each ReLU and compounded through depth; the κ_3 input and the per-neuron pattern (not the level) of the κ_4 diagonal are the decisive missing inputs; and a parity theorem shows why closures of old κ_3 built on even statistics lose it entirely. A Hermite ceiling ladder shows that second-order κ_4 carriers are capped near pattern correlation 0.5 while cubic births reach 0.8; the cubic-star feature built from it graded −4.4 %.

This write-up refers to submission #337457 (graded in Phase 2): adjusted score **2.7858e-9**, raw final-layer MSE **1.8592e-8**, util 0.150 (FLOPs as a fraction of the budget, §2.1). It is LordProcciuto's public e30_a chain (MIT; 3.1163e-9 as published, #335981) with kai_leal_gonzalez's (3,1) closure (MIT), plus our κ_4-pattern corrector (V55/V56, §5), our settings, and our refit and extension of e30_a's mean-correction table (§6.5). §2.3 separates their parts from ours. *For reference only:* our best result without e30_a's base chain is #337049 (e299safe, 3.2529e-9): our V29-based chain #334870 (v56As_fs, 3.6521e-9), on which §3–§5 are measured, plus mdhamidhosen's calibration and cost layer (§6.4) and e30_a's exact cost savings, fixed old basis and C_pre compaction with kai_leal_gonzalez's closure at mix 0.25 (§6.5).

21 October 2026. This is the main write-up. The full technical report is attached as the Appendix (full technical report), and "App. §x" refers to its sections. Each number and claim is labelled by its source:

- *graded: an official grade, with its submission id;*
- *measured: a run of our own scripts, with the result file named in App. §8.3;*
- *derived: algebra or arithmetic, with the derivation in the Appendix;*
- *estimate or untested: not yet measured;*
- *conjecture: an interpretation we have not tested, marked as such.*

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1. Summary

We started from 504aldo's public, MIT-licensed factorised third-cumulant chain ("V29"), which graded 5.418e-9 when submitted verbatim. Our changes take it to **3.6521e-9 (−32.6 %)**: raw MSE falls by 8.6 % (2.133e-8 → 1.950e-8) and util by 26 % (0.254 → 0.187) (derived). A final calibration and cost layer by mdhamidhosen on this chain (E282–E299, §6.4) takes it to **3.3946e-9 (#336377, −7.0 %)**; that layer is fitted and engineering, not new understanding. The submission this write-up refers to, **#337457 (2.7858e-9)**, moves our corrector onto LordProcciuto's cheaper public chain e30_a (MIT, 3.1163e-9 as published) with kai_leal_gonzalez's (3,1) closure; our corrector, our settings and our refit and extension of e30_a's correction table give −10.6 % there, −6.3 % of it from the fitted table (§2.3, §6.5). Of the contributions below, the V56 star (item 3) and the analysis behind it (items 1 and 3) act in #337457; the spherical identity and the parity theorem belong to our own chain.

Mechanistic contributions. These are what we consider our contribution to the understanding of white-box estimation:

1. **An error map with channel oracles** (§4; measured on $6{\cdot}10^7$-sample Monte-Carlo teachers). The closures are accurate when given true inputs; the error is input error made at each ReLU and accumulated through depth; the teacher's κ_3 and κ_4 inputs together would take the raw MSE to ×0.272 on MLP #0; the κ_4 gain lives in the **per-neuron pattern** of the κ_4 diagonal, not its level.
2. **A parity theorem** (§3.2; derived, verified numerically): old κ_3 is odd in the later weights, so every closure built on even statistics loses it.
3. **A ceiling ladder and the Hermite cubic-star feature V56** (§4.4, §5.2; identity derived and verified numerically to ≤ 8.1e-16; graded −4.4 %): second-order births cap the κ_4 pattern at 0.45–0.55, cubic births reach 0.80–0.85.
4. **An analysis of the κ_3-dressed C·T channel** (§4.5; oracle and component level, not graded): with the star it reaches the cubic-birth ceiling, and the chain's own legs reproduce it, but not yet at an affordable cost.
5. **Closure-input errors cancel** (§7.1 (2); measured at one step): the sum of their norms is 1.9–2.6× the norm of their sum (var against d3, cosine down to −0.83), so correcting one channel alone hurts.

6. **A spherical-input identity** (§3.1; derived, injection check measured, graded −1.1 % raw at zero cost).
7. **A negative-result atlas** (§7): 24 of our 60+ killed architecture and representation ideas, each with its gate and measured number, and three working conclusions that we label as empirical, not proven.

Engineering and fitted components. We state these plainly. Most of the gain from 5.418e-9 to 4.0784e-9 is **cost engineering**: Strassen placement, a transport join, dead-neuron compaction, skipping D21 (κ_3 slice, §2.2) entries of dead neurons, and an operation budget for the grader's hidden limits (§6). This is performance optimisation, not new understanding. The κ_4 gain from 4.0784e-9 to 3.6521e-9 (−10.5 %) passes through **V55, a linear map fitted offline** to our own Monte-Carlo teacher (§5.1); the V56 star enters as one more feature of that fitted map. Five constants are **tuned**, not derived: the feedback scale FB = 1.25 (§3.3), the adaptive-λ exponent β = 1.2, a ×1.25 on the V55 map's s3sq coefficient, the dense-age cut AGE_OLD = 5 and the old-basis rank R_OLD = 352. The −7.0 % of §6.4 is a Gauss-Newton calibration fitted on public MLPs plus mostly exact cost rewrites and two threshold approximations. The final −17.9 % (3.3946e-9 → 2.7858e-9, §6.5) comes from moving our corrector onto LordProcciuto's public chain, settings, and a refitted mean-correction table, which is a fitted component.

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2. Starting point: the reference method (not claimed)

2.1 Task and score

For a bias-free ReLU MLP of width n = 1024 and depth 16, with He-initialised weights and input $x \sim N(0, I)$, the estimator returns $E[h^{(l)}]$ for every layer. The score is

score = MSE_final × max(0.1, FLOPs / 2^{41}),

with the final-layer MSE measured against a 10^9-sample Monte-Carlo reference. FLOPs are metered by flopscope. We use **1 u = 2^{31} FLOPs**, about one dense 1024^3 product; the budget is then 1024 u, "util" is FLOPs/budget, and the 0.1 floor is 102.4 u. The grader caps each MLP at 120 s of wall time, 0.4 s of residual Python time outside metered operations, and 8 GB of address space. A capped MLP scores about 0.35, so a single cap in a hundred MLPs ruins a submission.

2.2 504aldo's V29 (MIT)

Our estimator modifies **504aldo's public V29** (`whest-p2-cumulant-k3`, forum post #18218), which implements the factorised K = 3 "simple" cumulant propagation of Wu et al. (arXiv:2605.05179, §4.3). Per layer it carries the mean, the covariance C, and the third cumulant κ_3 as a **sum of sources**, one per ReLU. Each source is stored as a propagator leg P_s, a birth leg A_s = P_s a_b, and thin n×16 feedback legs. The next ReLU needs two κ_3 slices, the diagonal d3 (d3_i = κ_iii) and the (2,1) slice D21 (D21_ic = κ_iic), read per source from contractions of Hadamard products of the legs. Sources older than four layers (the age cut AGE_OLD) live in a shared basis of rank R_OLD = 384 (the oldest in a nested rank-224 sub-basis). κ_4 is not carried: its off-diagonal part is regenerated each layer as G_off = λ_l C_off (the "λ law"), with a per-MLP adaptive λ. Products use pooled Strassen–Winograd.

The author prices V29 at 260 u and reports a tolerance law: extra final MSE ≈ 4.2e-6·ε^2, where ε is the relative RMS error of D21, so D21 must be right to about 2 %.

Graded baseline. V29, submitted verbatim, graded **5.418e-9** (raw 2.133e-8, util 0.254; **#331681**), matching the author's own submission.

Licence and scope of our claims. Every file we ship keeps 504aldo's MIT copyright notice and LICENSE, and carries a lineage header that names each change and its date. **Only the changes in §3–§6 are claimed as ours.** The sum-of-sources representation, the shared basis, the memoryless κ_4 law, the feedback legs and the pooled Strassen engine are the reference method.

2.3 The base of #337457: LordProcciuto's e30_a and kai_leal_gonzalez's closure (MIT; not claimed)

LordProcciuto's public **e30_a** (github.com/siming767-creator/whestbench-lordprocciuto; 3.1163e-9, #335981, cited) descends from V29. #337457 also uses **kai_leal_gonzalez's two-term closure of the (3,1) slice of κ_4** (github.com/al07-alt/whest-p2-k31-two-term; forum topic 18246).

Component of #337457	Origin
Sum-of-sources chain, shared basis, λ law, feedback legs, pooled Strassen	504aldo (V29), via e30_a
Removal of neurons with $\alpha < -2.85$, fixed-basis old level (rank 416/224) outside hot rows, Winograd leaf, exact fixed-cost savings, λ tilt, near-dead column cut τ_B (value 2.4 set by us)	LordProcciuto
Mean-correction table: 87 ridge features fitted on 70 public nets	LordProcciuto
Two-term (3,1) closure and its per-layer coefficients	kai_leal_gonzalez
κ_4-pattern corrector: V55 map and V56 cubic star (§5), coefficients refitted in mdhamidhosen's E-series (§6.4)	**ours**
Corrector scale 0.4, star/local split 0.7/0.2, closure mix 0.4, D21 ×0.98 at $L \geq 9$	**ours** (tuned)
Table refitted on our trajectory with 77 new features (164 in all; §6.5)	**ours** (fitted)

Not in #337457: V31 (§3.1; worse on e30_a, #337221), FB 1.25, our cost engineering (§6.1–§6.3), the E-series groups and rider. From e30_a's 3.1163e-9: corrector −2.1 %, settings −1.8 %, table −6.3 %, τ_B −0.8 % (−10.6 % in all; derived).

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3. Exact identities and derivations

3.1 Spherical-input identity (derived; V31)

Write the Gaussian input as $X = (R/\sqrt{n})\cdot Y$, with Y uniform on the sphere of radius $\sqrt{n}$ and $R \sim \chi_n$ independent of Y. A bias-free ReLU network is positively homogeneous of degree 1, so

$E[h(X)] = a_1 E[h(Y)]$, $a_1 = \sqrt{(2/n)}\,\Gamma((n+1)/2)/\Gamma(n/2) = 0.99975589$.

The sphere input has $C = I$ and $\kappa_3 = 0$, and its κ_4 is an exact harmonic core, $g = -6/(n+2)$ in the chain's representation. The estimator injects that core at layer 0, runs the existing κ_4 machinery unchanged, and rescales the output by a_1.

- **Measured:** the injection reproduces the predicted scale shift to 0.3 %.
- **Graded:** −1.1 % raw at zero cost (#331787).

The idea came from barnobarno666's public Phase-2 repository (idea only), pointed out to us by GPT-5.6 Sol in an outside consultation (App. §2.1). The derivation for this chain and its integration are ours.

3.2 Parity theorem: no mean-field part of the deep κ_3 (derived and verified; App. §2.2)

Take a source born at layer b. At any later layer l its legs are $P = W^{T(l)}\,D\,P'$ and $A = P\,a_b$, where P', a_b and the birth coefficients do not depend on $W^{(l)}$. Every quantity linear in the source's κ_3 (the slice products and d3) is therefore **odd** in $W^{(l)}$, while $C^{(l)}$, the leg Grams and the per-neuron variances are **even**. Because the weight ensemble is sign-symmetric,

$E[\,H_s^{(l)} \mid C^{(l)}, \mathrm{Gram}(A_s, P_s), \text{all state before } l\,] = 0$, exactly.

Consequences:

- Any closure that replaces old sources by a function of even statistics (mean-field closures, scale mixtures, polynomials in C) discards their entire contribution. This is consistent with the failures of such closures reported in 504aldo's public log (9.1e-7 and 2.3e-6) and with our learned residual correctors, which recover only 0.3–0.7 % of the residual (§7).
- Replacing a source by its conditional expectation given one fresh weight row has a relative-error floor of $\sqrt{(6/(r+8))}$ in the retained dimension r: 12 % at $r = 384$, far above the 2 % tolerance.
- κ_4 terms built from two κ_3 factors are even, and $C \otimes C$ is the only even invariant available. We read this as the reason the regenerated law $G_{off} = \lambda_l C_{off}$ works (conjecture beyond the derivation; not tested separately).

Status: derived, then verified (11 Oct; `experiments/probes/parity_check.py`). On the frozen V56 chain (width 128), flipping $W^{(l)}$ leaves every earlier layer and $C^{(l)}$, $\mathrm{var}^{(l)}$ bit-identical and flips μ, d3 and D21 exactly (difference 0.0; tested at $l = 5, 10$). The floor's algebra holds by Monte Carlo, and the bound $\|\tau\|^2 \leq (r+2)\|T\|^2/3$ is tight (it equals the exact operator norm) (eigen-computation, $r = 2$–8); a written proof is in App. §2.2. The floor is a statement about the idealised fresh-row

model. The mean $\mu^{(1)}$ is also odd, so closures that condition on μ are not covered. The argument originated in a consultation with Fable 5, an Anthropic Claude model run as a Claude Code subagent.

3.3 A half-carried feedback term (derived and measured; shipped as tuned FB 1.25)

Expanding the post-ReLU κ_3 to first order in C shows that the newborn source carries exactly **half** of two all-distinct K×C families (derived, App. §2.4). Write FB for the scale on these carried families: FB = 1 is V29, FB = 2 restores them in full. Regression on the slice residual gives coefficients of −0.55 to −0.9, against −1/2 predicted: consistent in sign and size, not exact.

Restoring the missing families in full makes the final mean worse: +3.4 % / +3.1 % on the teacher pair (MLP #0 and a synthetic net, §4), and ×1.024 on 100 MLPs at FB 2.0. A quarter restoration (FB 1.25) with the adaptive-λ exponent β = 1.2 scored ×0.9946 ± 0.0007 on the 100-MLP harness and graded 5.1284e-9 (#332110), at zero FLOPs. The shipped scale is therefore a **tuned constant**, not the derived one. Why the full restoration hurts is open; one explanation is ruled out (App. §6.5 (b)): the quadratic feedback cross term that also grows with FB is inert.

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4. The error map: where the chain's error lives

All results in this section are measured, using our Monte-Carlo teachers with $6{\cdot}10^7$ samples on public MLP #0 and a synthetic 1024×16 He network. They record the per-layer pre- and post-ReLU means, covariances, κ_3 slices and κ_4 diagonal. An **oracle** replaces one quantity inside the chain by the teacher's value and reruns the chain end to end. Two networks are a small sample; we state each result for the network it was measured on.

4.1 The closures are accurate; the error is accumulated input error

- **Mean closure.** Given the teacher's true marginals (μ, var, κ_{3iii}, κ_{4iiii}), the Edgeworth mean closure matches the teacher's ReLU mean to 5e-11–1.5e-10 at every layer.
- **Covariance closure.** Teacher-forced at one layer, the bivariate covariance closure falls to the teacher's noise.
- **Accumulation.** The chain's per-layer MSE grows smoothly, from 0.7e-9 at L1 to 2.29e-8 at L16.
- **Inherited share.** Forcing the true mean at every layer removes about 80 % of the final error.

We read these as: the error is **fresh input error** made at each ReLU and carried forward through depth.

4.2 Channel and joint oracles

The base is the plain chain at util 0.2476 (App. §4.8 gives the full table).

Teacher input injected (every layer)	MLP #0	Synthetic net
None (plain chain)	2.273e-8 (×1)	1.920e-8 (×1)
κ_3 slice + d3	×0.607	×0.716
κ_4 diagonal	×0.686	×0.885
κ_3 slice + d3 + κ_4 diagonal	**×0.272**	**×0.569**
κ_3 slice + d3 + κ_4 *pattern* at the chain's level	×0.294	×0.606

- **κ_3 is the larger single lever** on both networks.
- **The two inputs compound more than multiplicatively.** The product of the single-input ratios would be 0.416 and 0.634; jointly they give 0.272 and 0.569.
- **The remainder is consistent with the variance channel.** In the separate channel oracle of App. §4.2 (base 2.290e-8), the true variance, κ_3 diagonal and κ_4 diagonal together take MLP #0 to ×0.057, the truth's own noise floor. The two oracles inject different κ_3 quantities, so this is a consistency check, not a decomposition.

4.3 κ_3: where the error is real, and where the gain lives

We compared the chain's κ_3 with the teacher at every layer, and subtracted in quadrature a Gaussian estimate of the teacher's own sampling noise (derived; this assumes the two are independent):

Layer	D21 off-diagonal error	Teacher noise (est.)	Net D21 error	d3 norm ratio, chain/teacher
2	4.4 %	5.3 %	≈ 0	1.002
4	4.2 %	2.5 %	3.4 %	0.998
8	4.9 %	1.2 %	4.8 %	0.996
12	5.9 %	0.7 %	5.9 %	0.993
16	6.9 %	0.5 %	6.9 %	0.986

Under that noise estimate, the early "error" is mostly teacher noise, while the late slice error is real and grows with depth; the d3 level shrinks by 1.4 % at L16. Of the slice error, 91–96 % sits in the **all-distinct** post-ReLU entries $\kappa_3(h_i, h_j, h_k)$ (App. §4.5).

Localisation. Injecting the teacher's κ_3 only in a window of layers (MLP #0, base 2.2732e-8):

Window	Slice + d3	Slice only	d3 only
L1–4	×1.004	—	—
L5–8	×0.942	—	—
L9–12	×0.857	—	—
L13–16	×0.806	—	—
L1–8	—	×0.937	×1.012
L9–16	—	×0.834	×0.853

The gain lives in **L9–16**, where the slice and d3 matter about equally. Injecting the teacher's κ_3 into the feedback legs alone is worth ×1.001.

4.4 κ_4: the pattern, not the level

Injecting the teacher's κ_4 diagonal into the mean closure takes the raw MSE from 2.290e-8 to 1.609e-8. How that gain splits:

- **The teacher's pattern at the chain's own level** gives 1.551e-8, which is **108 %** of the gain.
- **The chain's pattern at the teacher's level** is *worse than no oracle* (+21.5 %).
- **In our oracles, κ_4 helps only through the mean closure's diagonal input.** True κ_4 everywhere except the mean closure gives ≈ ×1; the off-diagonal slices injected alone give ×1.03–1.22.
- **The chain's pattern is weak where it matters.** Over all neurons the chain's κ_4 diagonal correlates 0.78–0.83 with the truth; on the decisive neurons only 0.26–0.34. (*Pattern correlation* is the across-neuron correlation of predicted and teacher κ_4/v^2 on *decisive* neurons, $|\mu/\sigma| < 1.5$, which drive the mean closure.)

A ceiling ladder for κ_4 carriers. We measured the ceiling of whole *families* of birth laws by restarting a Gaussian h layers back and expanding each ReLU to Hermite order K (App. §5.4, $N = 4 \cdot 10^6$):

- **Quadratic births are capped in this ceiling model:** births that stop at second order reach pattern correlation 0.45–0.55. Every second-order carrier we and outside models proposed stalled in, near or below this band (−0.35 to 0.56, §7), consistent with the cap being the cause, though not a proof of it.
- **Cubic births suffice:** cubic births over six layers reach 0.80–0.85, and K = 4 adds ≤ 0.02.
- **Ages are complementary:** ages 1–6 are needed jointly. Ages 1–4 give 0.68–0.74, and ages 4–6 alone give only 0.37–0.41.

This ladder is why V56 uses a *cubic* feature.

4.5 The κ_3-dressed C·T channel (measured at oracle and component level; not graded)

The V56 star (§5.2) is the *Gaussian* cubic birth. A κ_3-dressed C·T readout, $ct_i = 24\, \Sigma_a\, g_a\, \bar{Y}_ai\, X_ai\, Q_ai$, reads the chain's κ_3 through the quadratic and cubic Hermite vertices. Here a runs over the neurons ("hubs") of a vertex layer, g_a is the ReLU's quadratic Hermite coefficient there, $\bar{Y}$ transports the hub to the target neuron i, X = C·(transport) is the covariance leg, and Q contracts the chain's κ_3 with two transports (App. §5.8; design from a 9 Oct Fable 5 consultation; every number below is ours):

- **Ceiling.** Sample-exact star + C·T reaches 0.82–0.85, matching the cubic-birth ceiling of §4.4. C·T alone sits at 0.42–0.44, where an earlier hub-pair carrier was killed; it is complementary to the star.
- **The chain's own legs reproduce it** at fidelity 0.993–0.996 on decisive neurons (component test).
- **End to end under an oracle map** (fitted to the teacher on the evaluation network, FLOPs not metered): ×0.773 with the sample-exact feature, ×0.819 with the chain's exact C·T on the available source subset.
- **Cost.** An exact age-collapse identity, Γ(a) = G(a) + Ystep(a) Γ(a+1), plus grouping by dense anchor, reduces it to two dense products per source and target (derived; matches the per-age route to 2.6–3.2e-7). At r = 64 a sketch keeps ×0.865 at about 40 u (estimate), roughly score-neutral. A rank-32 sketch keeps ×0.874 at an estimated 18–20 u. It is untested on V56 and is our main open lead.

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5. The κ_4-pattern programme (graded)

5.1 V55: a fitted, zero-FLOP κ_4-pattern map (fitted component)

Given §4.4, V55 corrects the pattern, not the level. At every layer ≥ 6 (≥ 4 from #334762 on), a **fitted linear map** turns seven per-neuron quantities the chain already holds into a predicted pattern of κ_4/v^2: s3sq = $(d3/v\text{^}(3/2))^2$; the row and column energies of the standardised D21; the correlation-row energy of C_pre; $|\mu/\sigma|$; the chain's own κ_4/v^2; and a constant. The pattern is rescaled to the chain's layer-mean level and fed **only** to the mean closure. The map is fitted offline, on decisive neurons, against our own teacher for MLP #0; the synthetic net is held out. It costs 0 u and +310 operations.

- **Pattern correlation:** 0.41–0.55 on MLP #0 against 0.24–0.41 for the chain, and 0.41–0.52 held out on the synthetic net.
- **100-MLP harness:** ×0.9455 [90 % CI 0.9410, 0.9498], with 97/100 MLPs improved.
- **Graded:** −5.4 % in both operation classes (A and B, §6.3). The harness projected 3.8764e-9 / 3.8562e-9; the grades were 3.8762e-9 (#334588) and 3.8582e-9 (#334591).

V55 is a regression on chain features, not a mechanism; its features saturate near pattern correlation 0.5 (App. §5.3).

5.2 V56: the Gaussian cubic-star feature

The cubic Hermite coefficient of the ReLU, c = $-\alpha\varphi(\alpha)/6$ in standardised units, together with the Wick identity $\kappa(u_a^3, u_b, u_c, u_d) = 6\rho_{ab}\,\rho_{ac}\,\rho_{ad}$, injects a **star**-shaped fourth cumulant downstream of every hub (derived; verified numerically to ≤ 8.1e-16, App. §2.5). V56 reads this star from the chain's **own** young dense source legs (ages ≤ 6), with the per-source coefficient $c_a = -\alpha_a\varphi(\alpha_a)/(6\sigma_a^2)$ recorded at birth (App. §5.8 notation):

$star_i = 24\,\Sigma_a\, c_a\, \bar{Y}_{ai}\, X_{ai}^3$.

The star enters as an eighth feature of the V55 map, at zero added FLOPs: util 0.1857, against 0.1855 for the V55 reference. Alone its ceiling is 0.47–0.52, but it is complementary to the V55 features: with teacher μ and C, leave-one-layer-out fits rise from 0.48–0.56 to 0.65–0.74. The star idea came from an outside-AI proposal, the "Hermite Tree Quartic Bank" (Round 18; unsigned), built on our own Hermite-family ceilings (App. §5.4–§5.5).

Graded. −4.4 % (#334805, B) and 3.6597e-9 (#334811, A). Refitting the map on the synthetic net gave a further −0.29 % / −0.21 %, reaching **3.6723e-9 (#334869) and 3.6521e-9 (#334870)**.

The κ_4 programme as a whole took the score from 4.0784e-9 to 3.6521e-9 (−10.5 %) at unchanged cost. Its coarse tuning is closed: scaling the star by 0.75 or 1.5 scored ×1.011, and limiting it to ages ≤ 4 scored ×1.0045; a 16 Oct sweep found only small further gains (K4C blend 1.1, ×0.995, graded #336382; App. §3.8).

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6. Engineering components: cost under the meter and the grader's limits

These sections are performance engineering. We report them because they carry most of the cost reduction and because the grader's hidden limits shaped every later design, not as contributions to understanding.

6.1 Graded trajectory

Step	Change (origin)	Score (id)	Raw MSE	Util
V29 verbatim (reference)	—	5.418e-9 (#331681)	2.133e-8	0.254
Strassen recursion floor 32 → 8; full depth on first call	Two more Strassen levels on the basis families	5.220e-9 (#331686); 5.195e-9 (#331732)	2.132e-8	0.2436
+ V31 spherical-input identity	§3.1, 0 FLOPs	5.137e-9 (#331787)	2.108e-8	0.2437
+ FB 1.25, β 1.2	§3.3 (tuned), 0 FLOPs	5.1284e-9 (#332110)	2.1046e-8	—
Transport join	Shared basis rebuilt at a join by transport, with no range finder	4.9519e-9 (#332257)	2.1788e-8	0.2273
+ AGE_OLD 5, R_FB 4	One more dense young age; feedback rank 16 → 4	4.8066e-9 (#332324)	2.1371e-8	0.2249
v48–v51 compaction	Dead neurons (ReLU slope Φ < 1e-2) leave the next transport; basis-side Strassen, aliased Grams; dead D21 columns and rows skipped (#332391, #332401, #332404, #332418; §6.2)	4.0902e-9 (#332420, A)	2.1636e-8	0.189
v52 operation budget; R_OLD 352 and exact trims	§6.3; old rank, QR-derived transport, feedback rank 2	B 4.0995e-9 (#332626); A 4.0784e-9 (#332621)	2.197e-8 (B)	—
V55 κ_4-pattern map; s3sq ×1.25, B / A	§5.1 (fitted; coefficient tuned)	3.8539e-9 (#334762) / 3.8331e-9 (#334763)	—	—
V56 cubic star, B / A	§5.2	3.6829e-9 (#334805) / 3.6597e-9 (#334811)	1.9706e-8 / 1.9539e-8	—
+ map fitted on the synthetic net, B / A		3.6723e-9 (#334869) / **3.6521e-9 (#334870)**	1.950e-8 (A)	0.187
E282–E299 (mdhamidhosen): 24-group calibration, rider, corrector refits, exact cost rewrites, thresholds	§6.4 (fitted; engineering)	3.5890e-9 (#335603) → 3.5085e-9 (#335729) → 3.3946e-9 (#336377)	1.8824e-8	0.180
+ e30_a's exact cost savings (LordProcciuto); kai_leal_gonzalez's (3,1) closure at mix 0.25	§6.5	3.3325e-9 (#336757)	1.8742e-8	0.178
+ e30_a's fixed old basis and C_pre compaction (LordProcciuto)	§6.5	3.2529e-9 (#337049)	1.9107e-8	0.170
Our corrector on LordProcciuto's e30_a (scale 0.4); kai_leal_gonzalez's (3,1) closure	§2.3, §6.5	3.0302e-9 (#337175)	1.9953e-8	0.152
+ D21 ×0.98, closure mix 0.4, corrector star/local split	§6.5 (tuned)	2.9956e-9 (#337258)	1.9723e-8	0.152
+ e30_a's 87-feature table refitted on our trajectory, + star modulators (N2)	§6.5 (fitted)	2.8475e-9 (#337301)	1.8747e-8	0.152
+ e30_a's near-dead column cut, τ_B set to 2.4	§6.5	2.8256e-9 (#337336)	1.8869e-8	0.150
+ table features: star^2, d3·star, corrector-local (N3, 122)	§6.5 (fitted)	2.7975e-9 (#337368)	1.8681e-8	0.150
+ D21/C row energies, carried delta and star (N4a, 164)	§6.5 (fitted)	**2.7858e-9 (#337457)**	**1.8592e-8**	**0.150**

Util is the grader's figure or adjusted ÷ raw. "—" means the value is not listed here. The B class keeps ≤ ~18k warm-call operations for a safety margin under the residual-time cap; the A class runs at about 20k.

6.2 Cost levers

The levers of the table: the transport join (−3.4 %); AGE_OLD 5 with feedback rank 4 (−2.9 %; on V56 the five nearby age/rank settings score ×1.026–1.104, App. §6.6 (c)); input-side dead-neuron compaction with skipped D21 columns and rows (−3.7 %, −3.7 %, −3.4 %; compacting the output side failed in every form, ×1.6 to ×10); Strassen floor 32 → 8 (−3.7 %); and basis-side Strassen with joiner Grams written as YY^T, billed at half (−4.6 %). Strassen level 6 exceeded 8 GB. Over the cost programme the trims removed 22 % of the metered FLOPs (util 0.2437 → 0.189) for +2.8 % raw.

6.3 The residual-time cap and the operation budget

Two variants that ran cleanly on our machines were graded catastrophically (#332402 at 0.0434, #332425 at 0.0204: about 12 % and 6 % of MLPs capped). The residual Python time tracks the number of metered operations (0.053–0.059 ms per op on Kaggle for every variant); the failing variants issued 21.7–22.3k operations. Our fix is an **operation budget**: Strassen only where it saves at least about 2 u per 1,000 operations. Class **B** (wider safety margin) keeps ≤ ∼18k operations, class **A** about 20k; neither has hit the residual-time cap since (#332566, #332567). A second hidden limit (an address-space cap) failed four 3-D forms of the V56 star (#334786–#334799) that passed in-process; `whest validate` catches neither limit, so our pre-submission check is an operation count plus a subprocess run.

6.4 Final calibration and cost layer (E282–E299, mdhamidhosen; fitted and engineering)

From 12 Oct, mdhamidhosen built on #334870 (numbers from his ledger, re-checked against the grader; App. §3.7):

- **Calibration (fitted).** A Gauss-Newton fit of 24 per-layer calibration groups (scales of the closure inputs and λ) on public mini MLPs 0–47 gave −1.66 % ± 0.20 % on unseen MLPs 48–99 (E282); a cubic final-layer ridge rider (900 full-split MLPs) and four end-to-end refits of the V56 κ_4 corrector followed (−0.77 / −0.48 / −0.33 / −0.37 % unseen; E291). Further steps gave ≤ 0.08 %.
- **Exact cost rewrites (engineering).** Grouped PK2 einsums (graded util −0.77 %), Strassen floor 4 and Strassen on the nested projections (−0.3 %, −0.48 % FLOPs in his ledger), and the Strassen-Winograd kernel (−0.77 % FLOPs) leave predictions unchanged up to rounding. With the D21 and dead-neuron thresholds raised (approximations; −0.73 % graded, E296 → E298) they give **3.3946e-9 (#336377, E299)**.

6.5 Final days: our corrector on a public chain, and a refitted table (17–21 Oct; measured and graded)

After the 17 Oct extension (grader on flopscope 0.12.3, which bills our chains identically on 100 Kaggle MLPs), we reviewed the new public write-ups and code and measured the candidates on the 100-MLP harness before grading; two (e299safe and V31 on e30_a) were checked on MLP #0 only (App. §3.9).

- **Exact cost savings from e30_a in our chain:** Gram-core skipping, R-factor rotation, folded constants: −1.4 % FLOPs, predictions unchanged up to rounding (#336711, #336719). kai_leal_gonzalez's two-term (3,1) closure (MIT, forum write-up) hurt at full strength (#336697, +0.7 %) but helped at mix 0.25 (#336757, −0.4 %). e30_a's fixed old basis and C_pre compaction gave our own-lineage best (#337049, 3.2529e-9).
- **Our corrector on LordProcciuto's public e30_a chain** (MIT; 158.8 u against 184.1 u for our e299xrk in the warm call on MLP #0, call-site ledger): at full strength it hurt (×1.085, MLP #0); at scale 0.3–0.4 it gave ×0.975–0.977 on 100 MLPs (94/100 at 0.3), graded ×0.979 against e30_a's published 3.1163e-9 (#336774). The gain comes from the V56 star (at scale 0.3: ×0.9768 with it, ×1.0024 without it); the local features duplicate e30_a's 87-feature mean-correction table.
- **Refitting e30_a's table on our trajectory (fitted).** The table was fitted on e30_a's own uncorrected trajectory. Refitted by ridge regression (grouped 5-fold CV) on teacher-forced dumps of public mini nets 0–69, with star modulators and the corrector's own κ_4 change added (N2, 101 features), it gave ×0.930 on held-out nets 70–99 (30/30) and graded −4.9 % (#337301). The original table transfers poorly to our trajectory, so the refit must follow larger changes of the chain. New non-local features paid further (N3: ×0.990 held-out, graded −1.0 %, #337368; N4a: ×0.9966 held-out, graded −0.42 %, #337457); fitting on 341 instead of 70 nets, an on-policy refit and the (3,1) orientation fix of forum topic 18246 did not (×0.9997, ×0.997–1.005, ×1.0005).
- **Measured and rejected:** V31 on e30_a (×0.985 on MLP #0, graded +3.9 %, #337221); e30_a's row compaction in our chain (×1.063, MLP #0); refitting the table after cheaper settings (rank 352: ×1.04 even with refit).
- **Operations:** the corrector adds about 0.8k warm operations to e30_a's 19.8k (20.6k on MLP #0, measured before the table extensions; A class); every e30_a-based submission graded with no capped MLP (a single cap would add about 3.5e-3 to the score).

- **Rules note:** the organisers allow building on public MIT code with attribution if the team adds substantial work of its own (forum topic 18240). In #337457 our own contribution is the corrector (V55/V56), the closure and corrector settings and the refitted, extended table; the base chain is LordProcciuto's e30_a and the (3,1) closure is kai_leal_gonzalez's (§2.3).

7. Negative-result atlas

"Gate" is the pre-registered pass criterion. Kill types: C, a ceiling or oracle (an upper bound); P, a teacher-forced component test; E, an end-to-end prototype; K, a knob on the 100-MLP harness. Ratios are final-MSE or score ratios against the chain. Details are in App. §6.

Idea	Gate	Measured	Kill
Compensating closure with no old tier (8 models proposed variants)	≤ chain MSE	7.43e-7, 32× the chain; every knob 6.8–9.9e-7	C
Mean-field / even-statistic closures of old κ_3	—	Zero by the parity theorem: old κ_3 is odd in later weights (§3.2)	derived, verified
Low-rank Hadamard products of source legs	low rank	A∘P needs 681–778 of 1024 directions for 99 % energy	C
Query surrogates: trace, diagonal Hermite, TensorSketch	D21 ≤ 2 %	98.2 %, 95.4 %, 790 % at L14	C/P
Source-free dressed hypertree (DHMP)	next D21 ≤ 2.5 %	36–40 %	P
Hermite-tree κ_3 replacing the chain	closure ≤ chain	10–20× worse	P
Every second-order κ_4 carrier (8 designs)	pattern corr ≥ 0.70	−0.35 to 0.56	C
Learned residual correctors (ridge, GBM; 102,400 neurons)	—	0.3–0.7 % of the residual recovered	E
Monte-Carlo control variates	—	Tangent-surrogate corr 0.15–0.29 at L16; variance reduction 1.0–2.2×	P
C·T channel via age-collapsed sketches (ACDS), r = 64	score gain at ≤ ∼30 u	Oracle E2E ×0.865 at ≈ 40 u (estimate): score about neutral. r = 32 (×0.874, est. 18–20 u) is open	E (parked)
Missing cubic-response C×κ_3 term (RLL)	ρ ≥ 0.1 with the residual at L ≥ 9	ρ ≈ 0; adding half the term worsens the slice residual by 15–42 %	P
γ-patch: FB 2.0 with the quadratic cross term held	ratio < 1	×1.0243, the same as unpatched FB 2.0 (×1.0244)	K
$\kappa_4 \to \kappa_3$ feed from the exact κ_4 ($N = 10^6$)	exact ≤ 0.80 pass, > 0.92 kill	×1.115	C/E
κ_3 level rescaling, d3 or D21 ×0.98–1.02	< 1 on both nets	×0.984 to ×1.074; the one gain does not transfer	K
Single shared pool for the whole κ_3	D21 ≤ 2 % at some r ≤ 512	3.4–6.3 % at r = 512; 0.2–1 % only at r = 768	C
Deferred exact join (youngest source exact)	D21 ≤ 2 %	2.5–8 % at r = 352–512	C
Age/rank re-split of V56 (5 settings)	score ratio < 1	×1.026 to ×1.104	K
Checkpointed birth archive (κ_3 skipped in a window, sources kept)	≈ ×1	×14.8 (L1–4) to ×125.6 (L1–12); no recovery after the window	E
Trace projection of old κ_3 (Wu et al.)	D21 ≤ 2 %	26–42 %	C
Cost-first κ_3 representation at 60 % of the cost	2 % channel accuracy	None found; slice-only states are not exact to first order (derived)	derived
Network-conditioned routing over 86 archived variants	≥ 3 % gain	Oracle ceiling 0.80 % before overhead	C
Learned pair-memory student PM364 (source-free, 364 parameters)	MSE to V56 ≤ 1e-9; D21, d3 ≤ 2 %	MSE −0.9 % after training; D21 58–70 %, d3 25–39 %	E
Exact old tier (no confinement)	score gain	−3.9 % MSE for +19 % cost	E

Idea	Gate	Measured	Kill
Strassen level 6	≤ 8 GB	8.13 GB	cap

7.1 Working conclusions (empirical; not proven)

1. Within the sum-of-sources family, cost and accuracy traded roughly one for one in every knob we swept. This is an empirical regularity over our sweeps, not a theorem. On the 100-MLP harness the shipped constants sit at or within about 0.5 % of the score optimum of the settings tested (App. §3.8): β 1.2 (×0.9993), λ ×0.9 (×1.0026), the FB × β grid (within +0.12 %), R_OLD 448 (×1.0113) and AGE_OLD 6 on V56 (×1.014). On the five-point age/rank grid on V56, the best point gains 0.5 % raw for 3.2 % more util, and saving 4.3 % util costs 10.3 % raw. The exceptions in our records are levers that are **exact or nearly exact** (transport join, compaction, Strassen placement, aliased Grams); these are where our cost gains came from. A 27 Sep audit of 74 earlier kills revived 8, all cost or knob levers killed on the wrong criterion; no architecture kill was revived.

2. Local corrections of one channel made the final mean worse in every case we tested: full FB restoration (+3.4 % / +3.1 %); the γ-patch (×1.0243); the RLL term (slice residual 15–42 % worse); scaling the chain's $\kappa_4 \to \kappa_3$ feed in either direction (−5 % costs 0.45 %; +10 %, toward the larger exact feed, costs 9.4 %); injecting the sampled exact feed (×1.115); raising d3 toward the teacher's level (×1.032 / ×1.015); single κ_4 slices (×1.03–1.22). Only **joint** input oracles give large gains (×0.272 against ×0.607 and ×0.686 alone). *Measured at one step (App. §7.5):* the chain's fresh closure-input errors cancel. The sum of their norms is 1.9–2.6× the norm of their sum, chiefly var against d3 (cosine down to −0.83). Correcting var alone (L2–L9) or d3 alone (L4–L10) makes the step worse by up to ×1.31 and ×1.13, while var and d3 together always help (×0.56–0.79). How this carries through depth to the final mean remains a conjecture, and for the feed, Monte-Carlo noise is a competing explanation. The practical rule we adopted: local tests are necessary gates, not sufficient ones; candidates are promoted only on the end-to-end mean.

3. Every source-free or pooled κ_3 state we built failed its accuracy gate. *Conjecture:* the common cause is that the all-distinct κ_3 entries, where 91–96 % of the slice error sits (§4.3), are information a cheap state does not hold. Evidence:

- **The slice-state argument (derived in an outside consultation; App. §6.6 (g)).** A perturbation H = ε Sym(e_a, e_b, e_c) with distinct a, b, c is invisible to the mean, the covariance and the slice, yet after one transport it produces a slice and a diagonal that are first order in ε. A recursion on the slice alone therefore cannot be exact to first order, and a linear state closed under every transport that returns every diagonal needs $C(n+2, 3) \approx 1.8 \cdot 10^8$ numbers. The argument does **not** exclude nonlinear, restricted-family or approximate encodings (the source chain itself is one).
- **Age memory (measured).** Keeping only sources of age ≤ h leaves D21 errors of 84 / 65 / 46 / 31 / 16 % at h = 2 / 4 / 6 / 8 / 10; six age modes still leave 8.5 %.
- **Pooling (measured).** Pooling only the old sources costs 1.2–1.3 % at r = 384, but pooling everything needs r = 768.
- **Skipping and learning (measured).** κ_3 could not be deferred to a checkpoint (×14.8 to ×125.6), and our per-neuron-feature student PM364 did not learn it: its D21 error at L2, the first transport, is 57.6 % and does not move under training (two seeds). This rules out that student design, not learned states in general.
- **Averaging (derived).** By the parity theorem (§3.2), closures built on even statistics lose old κ_3 entirely.

Where our cost goes. A V56 call-site ledger puts 193.8 u in one warm call: about 70 u in the D21 hub readout, 63 u in young-leg transport, 20 u in the old basis, and about 40 u elsewhere, which issues 11.3k of the 20.6k operations. The 133 u in the two per-source families grows with depth, and nothing we tested removes it without at least a comparable accuracy loss. For comparison, the reference algorithm of Wu et al. costs about 4,150 u at our size (derived from their cost table), about 21× V56.

Relation to the leaderboard. On the 10 Oct board the six highest entries have raw MSE 1.14–1.65e-8 at util 0.106–0.144. On 20–21 Oct our row shows 2.8e-9 at rank 34 (tied; the board rounds to two significant figures); the prize ranking comes from the organisers' private re-evaluation. One entry at rank 11 has V29-level raw MSE (2.10e-8) at util 0.102. At #337457's raw MSE (1.86e-8), util 0.10 would score about 1.9e-9 (derived). We do not know the leaders' mechanisms. Our measurements exclude only the specific κ_3 representations in the table above as cheaper drop-in replacements inside our chain.

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8. Reproducibility

Versions. whestbench 0.16.1, flopscope 0.12.1 (0.12.3 for the 17–21 Oct submissions, including #337457; identical billing on 100 MLPs, App. §3.9), dataset revision v2-phase2. Every graded number carries its AIcrowd submission id.

Path	Contents
`submissions/<variant>/estimator.py`	Every shipped estimator, with a lineage header naming each change: #337457 is `e30kc1n4/`, #334870 `v56As_fs/`, #334869 `v56s_fs/`, the reference `v29/`
`experiments/variants/e30k_x8.py` – `e30k_x14.py`, `kaggle/x8*`, `kaggle/x10` – `x14`, `external/`	The final-days line (§6.5): harness rounds, the self-contained Kaggle pipelines that dump, refit and evaluate the table, and copies of e30_a and the (3,1) closure (§2.3)
github.com/hamidhosen42/ARC-White-Box (private until release; available to the organisers on request)	mdhamidhosen's E-series (E299 = #336377), its ledgers and calibration tools
`experiments/variants/`, `experiments/probes/`, `experiments/student/`	Oracles (`r20_joint_oracle.py`, `r20_feed_oracle.py`, `v55_k4c.py`, `v56_star.py`), probes (`site_ledger.py`, `k3_error_profile.py`, `r17_hermite_family_ceiling.py`, `kaggle_pair_compare.py`), the PM364 student
`kaggle/`, `results/summaries/`, `consultations/`, `writeup/`	Kernel wrappers and teachers; result files; consultation notes and verdicts; this document, the Appendix and the PDF builder

App. §8.3 maps every Appendix section to its script and result file; known gaps in our records are in App. §8.4.

The 100-MLP harness. It runs the public mini split with its 10^9-sample truth on five Kaggle CPU runners. `kaggle_pair_compare.py` reports the paired geometric-mean ratio, wins, a bootstrap 90 % interval, the score and the residual-time profile. The base is bit-identical across runners, and paired standard errors are 0.01–0.15 %. Up to 16 Oct every predicted sign matched the grade; for single changes worth more than about 1 % the predicted size was within 0.05–0.7 percentage points; for gains under 0.5 % the size was unreliable. In the final days one non-significant joint prediction had the wrong sign (c1: ×1.0035 n.s., graded −1.1 %), and the refitted table N2 was over-predicted (×0.935, held-out ×0.930; graded ×0.951) (App. §3.9). The harness measures residual time but does not enforce the cap.

Building the PDFs. `python writeup/build_pdf.py` builds the Appendix; `python writeup/build_pdf.py writeup/WRITEUP_MAIN.md --out WRITEUP_MAIN --compact` builds this document. All scripts will be released under MIT after the competition closes.

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9. Disclosure of AI use

Public code we build on. #337457 runs on LordProcciuto's e30_a (MIT; its write-up discloses Claude use) and uses kai_leal_gonzalez's two-term (3,1) closure (MIT; its repository also discloses Claude use). e30_a's MIT notice (504aldo, LordProcciuto) is kept in the shipped files; kai_leal_gonzalez's closure is credited in the lineage header, and the released code adds his MIT notice. The corrector, the settings and the refitted, extended table are ours (§2.3, §6.5).

Code. Apart from the public code we build on (504aldo, LordProcciuto, kai_leal_gonzalez), essentially all code in our repository was written by AI coding assistants under the team's direction: every estimator variant, oracle, probe, harness runner and Kaggle pipeline (including the 17–21 Oct table dump-and-refit kernels) and the PM364 student. The assistants were Anthropic Claude models through Claude Code: mainly Claude Opus 5 and 5.5, with Fable 5, Opus and Sonnet subagents for specific derivations and measurements. The team chose what to build, ordered every run and reviewed the results. The new table features of §6.5 were proposed by Claude Code and selected by held-out measurement. The E-series layer of §6.4 (E282–E299) and its calibration fits were developed by mdhamidhosen in his own repository (§8); this statement covers our repository, and the E-series is covered by its author's own statement. Its graded numbers were re-checked against the grader.

Ideas. Many architecture proposals came from consultations with outside AI systems (GPT-5.6 (Sol, Terra), GPT Astra 6, Gemini 3.1 Pro, Grok, DeepSeek, Copilot, SciMaster/Bohrium, and several unsigned answers) over 20 numbered rounds (App. §6.4, §9). Of the shipped ideas:

- the spherical-input reparameterisation was pointed out in an outside consultation, which referred to a public repository;
- the cubic-star feature comes from an outside proposal built on our own ceilings;
- the parity argument and the FB-scale derivation came from Fable 5 subagent work with us, as did the κ_3-dressed C·T readout (§4.5) and the $\kappa_4 \to \kappa_3$ feed design (§7), 9–10 Oct;
- the slice-state argument (§7.1) came from an outside consultation.

Most were killed (§7); several outside documents reported measurements that no run supported (App. §6.4).

Write-up. Almost all of the text of this write-up and of the Appendix was drafted by Claude (Claude Code) from our result files and notes, and revised by the team. AI agents also reviewed the documents: a fact-check of every number against the result files (11 Oct) and a five-lane review of numbers, consistency, rules and attribution (21 Oct). The team checked each finding before accepting the edit and takes responsibility for every claim. (App. §9 lists the 19 Oct agent reviews.)

Validation. Apart from the cited figures of 504aldo, Wu et al. and the public leaderboard, and the numbers marked as estimates, every number here was produced by our own scripts (or derived from them) or by the official grader. Every outside claim was re-derived or re-measured before we used it. On 11 Oct 2026 an automated, AI-run fact-check compared every number then in this document with the result files; on 21 Oct a second review re-checked all 63 submission ids against the grader and about 165 measured numbers against their files. The errors found were corrected. No leaderboard feedback entered any fitted coefficient: the V55 map was fitted against our own teacher for MLP #0, the shipped V56 maps against our own teacher for the synthetic network, and the E-series calibration on public dataset MLPs with their published truth (mini 0–47, rider on full-split 100–999), validated on unseen mini MLPs 48–99; the refitted e30_a table (§6.5) on public mini nets 0–69, validated on unseen nets 70–99. Discrete choices (corrector scale, A or B class, which variant to keep) were made on the harness and, in a few cases, between graded submissions.

Understanding: what was verified numerically and what is only derived (several derivations were produced with AI assistance).

- *Verified numerically:* the cubic-star and Wick identities (to ≤ 8.1e-16) and their readout factors (to ≤ 9.0e-15); the spherical identity's scale shift (to 0.3 %); the age-collapse and anchor-grouping identities (to 2.6–3.2e-7); the closure accuracy given true inputs (§4.1); the implementation parity checks (PM364's flopscope and PyTorch versions agree to 2.8e-7); the parity theorem's sign flip on the real chain (exact) and the $\sqrt{(6/(r+8))}$ floor (Monte Carlo plus exact operator norm, with a written proof).
- *Derived, consistent with measurements but not exactly confirmed:* the half-carried feedback term (regression −0.55 to −0.9 against −1/2).
- *Derived only:* the slice-state argument and the C(n+2, 3) count, and the Wu et al. cost comparison.
- *Conjectures:* the depth propagation of the (one-step measured) error cancellation of §7.1 (2), the common cause in §7.1 (3), and the link from parity to the G_off = λ_l C_off law (§3.2).

References

1. Wilson Wu, Victor Lecomte, Michael Winer, George Robinson, Jacob Hilton, Paul Christiano. *Estimating the expected output of wide random MLPs more efficiently than sampling.* arXiv:2605.05179 (v2, 14 May 2026). The factorised K = 3 "simple" cumulant chain (§4.3), its cost table, and the reference implementations.
2. 504aldo, *whest-p2-cumulant-k3* (GitHub, MIT licence), and WhestBench Phase 2 forum post #18218. The V29 factorised third-cumulant chain used as our reference method.
3. WhestBench (whestbench 0.16.1) and flopscope (0.12.1), the ARC White-Box Estimation Challenge 2026 benchmark and FLOP meter (Alignment Research Center / AIcrowd). The organiser ruling of 11 Sep 2026 states that Strassen–Winograd is legal at any recursion depth.
4. Appendix (full technical report), Hack2Publish, 21 Oct 2026 (`writeup/ALGORITHM_TRACK_PAPER.md`).
5. LordProcciuto, *e30_a*, github.com/siming767-creator/whestbench-lordprocciuto (MIT), and Phase-2 write-up (graded #335981). The base chain of #337457.
6. kai_leal_gonzalez, two-term closure of the (3,1) slice of κ_4, github.com/al07-alt/whest-p2-k31-two-term (MIT), forum topic 18246.
7. barnobarno666, public Phase-2 repository (spherical-input reparameterisation).
8. Organiser ruling on building on other participants' public MIT code, forum topic 18240 (5 Oct 2026).